

Improved Project Alpha: Reproducibility & Data Integration (v3)

1. Environment Setup

Environment: macOS with Conda environments 'macs2' and 'base' activated as needed.

Core tools: sqlite3, bedtools, liftOver, awk, curl, bgzip, tabix.

Working directory: /Users/shaketaveal/Desktop/Improved Project Alpha

2. Directory Structure

Key directories:

- data/reference/chains : UCSC liftOver chain files - data/tmp : intermediate BEDs and lifted mappings - data/processed : processed enhancer sets per species - data/processed/conservation : cross-species conservation tables - data/processed/reports : final summaries and CSV reports

3. Species Processed

- Chicken (galGal6 → galGal7b validation via chain availability) - Dog (canFam3 → hg38/mm10, BarkBase ATAC + H3K27ac) - Pig (susScr11 → hg38/mm10, H3K27ac + ATAC)

4. LiftOver Mapping

All narrowPeak or BED4 enhancer inputs lifted to hg38 and mm10 via chain files: liftOver -bedPlus=4 [input] [chain] [output] [unmapped]

Mapping coordinates verified with wc -l and intersect counts using bedtools.

5. SQLite Schema & Integration

Database: ipa.db Key tables: - enhancers : canonical enhancer metadata - species : species identifiers and names - *_flags : tissue × species conservation flags - *_annotated : SQL views combining enhancers and conservation flags Views created: - chicken_[brain|heart|liver]_annotated - dog_[brain|heart|liver]_annotated - pig_[brain|heart|liver]_annotated

6. Reporting and Outputs

Final summary: conservation_summary.csv Columns: species, tissue, n_human, pct_human, n_mouse, pct_mouse, total Generated via SQL aggregation with COALESCE() to handle missing conservation flags.

Example result snippet:

```
species,tissue,n_human,pct_human,n_mouse,pct_mouse,total
dog,heart,26747,28.19,9150,9.64,94874 pig,heart,24,4.2,1,0.17,573
```

7. Quality Control Notes

- Low pig heart overlaps expected due to ATAC vs H3K27ac mismatch. - Chain completeness verified via UCSC md5 and successful coordinate mapping. - Data consistency validated through record counts pre- and post-import.

8. Next Milestones

Next analysis modules: 1. CTCF proximity: enhancer centers ± 1 kb intersect with `ctcf_sites` table. 2. GWAS enrichment: overlap human-mapped enhancers with `gwas_snps`. 3. Gene-level rollups: join `enhancer_gene_links` → summarize conservation per gene. 4. Visual QC: UCSC BED tracks compressed and indexed via `bgzip/tabix`.

Generated automatically for full reproducibility.